Critical Care / ICU OSCE Cases

Question and answer review by exam year, 2015 to 2025

Key: Drug names **Critical point** ★ Repeated case Answers follow Surviving Sepsis Campaign 2026, ASPEN/SCCM and KDIGO guidelines.

2025

Case 1: Hemorrhagic (Hypovolemic) Shock and Parenteral Nutrition

Case details: Gunshot wound to the abdomen with major blood loss. Received blood and fluids but still hypotensive, tachycardic and tachypneic; labs show metabolic acidosis. NPO and expected to stay NPO for 8 days. Weight 70 kg (given later in the stem).

Question	Answer
Q1. What will you give (class) to improve his condition?	• **Stop the bleeding first:** surgical or damage-control hemostasis is the definitive treatment • Blood products: balanced massive-transfusion ratio PRBC : plasma : platelets about 1 : 1 : 1; limit crystalloids • Tranexamic acid: 1 g IV over 10 min within 3 h of injury, then 1 g over 8 h (CRASH-2) • Calcium: replace ionized Ca during massive transfusion (Calcium chloride 1 g IV); keep the patient warm • Vasopressor (the class asked): only a bridge when still hypotensive after volume and blood • Permissive hypotension until bleeding control (SBP 80–90 mm Hg, no head injury); then MAP ≥65 mm Hg • Non-pharm: **damage-control surgery and hemorrhage control**, keep warm (avoid the lethal triad: hypothermia, acidosis, coagulopathy), limit crystalloids, avoid hemodilution • Monitor: MAP/BP and HR continuously, lactate and base deficit every 2–4 h, Hgb, INR, fibrinogen/TEG, platelets, ionized Ca, temperature, urine output ≥0.5 mL/kg/h
Q2. They decided to give vasopressors. Which agent?	• Norepinephrine: start 0.05 mcg/kg/min, titrate to MAP ≥65 mm Hg (peripheral line acceptable for a short time) • Add Vasopressin 0.03–0.04 units/min if the Norepinephrine requirement keeps rising • Do not treat hypotension with vasopressors alone: continued bleeding needs blood and surgery • Non-pharm: fluid/blood resuscitation first, bleeding control; insert an arterial line and a central line • Monitor: MAP every 5 min while titrating, arterial line, HR and ECG (arrhythmia), lactate, urine output, capillary refill, digital/limb perfusion, extravasation at the IV site
Q3. Why Norepinephrine and not Dopamine? What is the evidence?	• Dopamine causes more tachyarrhythmias (SOAP II: 24% vs 12%, mainly atrial fibrillation) • SOAP II: no overall mortality difference, but higher 28-day mortality with Dopamine in the cardiogenic shock subgroup • Norepinephrine: potent α1 (vasoconstriction) with mild β1; predictable response, less tachycardia • Surviving Sepsis Campaign 2021: Norepinephrine is the first-line vasopressor (strong recommendation)
Q4. What do the labs show? Metabolic acidosis?	• Expected: low pH, low bicarbonate, high anion gap, raised lactate (type A lactic acidosis) • Also look for: low base excess, raised SCr/BUN (hypoperfusion, AKI), falling Hgb, coagulopathy, low ionized Ca • Hyperchloremic component can be added by large volumes of 0.9% NaCl
Q5. Why did it happen?	• Hemorrhage → low oxygen delivery → anaerobic glycolysis → lactate and H+ accumulation • Poor renal perfusion/AKI reduces acid excretion • Large NS volumes add hyperchloremic acidosis • Treat the cause (restore perfusion); give bicarbonate only for severe acidemia (pH ≤7.2 with AKI) per SSC 2026 and BICAR-ICU
Q6. What is the compensatory mechanism?	• Respiratory: hyperventilation lowers PaCO2; expected PaCO2 = 1.5 × HCO3 + 8 (±2) (Winter formula) • Buffering: bicarbonate, proteins, bone (immediate) • Renal: more H+ excretion and bicarbonate regeneration (days), limited here by shock and AKI • A PaCO2 higher than expected = coexisting respiratory failure (ventilatory support)
Q7. Is the patient a candidate for TPN? Why?	• Yes, if enteral nutrition cannot be used (bowel injury, resection, ileus) and NPO is expected ≥7 days (here 8 days) • Try enteral feeding first (even trophic or jejunal) as soon as hemodynamically stable • ASPEN/SCCM: in a well-nourished patient, wait about 7 days before PN; start earlier if high nutrition risk (NUTRIC/NRS-2002) • Start PN at about 50% of goal and advance; correct K, Mg, Phos first (refeeding risk) • Monitor: glucose every 4–6 h; BMP, Mg, Phos daily for 3 days then 2–3 times weekly; TG and LFTs weekly; weight, fluid balance, line site; nutrition reassessment weekly
Q8. 70 kg, 14 kcal/kg/day: total calories?	• 70 kg × 14 kcal/kg = **980 kcal/day** • 14 kcal/kg is a hypocaloric (permissive underfeeding) target for the early phase; advance toward 25–30 kcal/kg when stable

Question	Answer
Q9. Protein 1.2 g/kg: grams and kcal?	• 70 × 1.2 = **84 g protein/day** • 84 g × 4 kcal/g = **336 kcal** from protein • Non-protein calories = 980 − 336 = 644 kcal • Example split: Dextrose 150 g (3.4 kcal/g = 510 kcal) + Lipid emulsion about 13 g (10 kcal/g = 134 kcal) • Check: glucose infusion rate = 150,000 mg ÷ 70 kg ÷ 1440 min = 1.5 mg/kg/min (limit 4–5)
Q10. Central line removed; he continues peripheral PN. Maximum osmolarity? Why?	• Maximum **900 mOsm/L** for peripheral PN (ASPEN, INS) • Why: hyperosmolar solutions injure the vein wall → phlebitis, thrombosis, extravasation injury • Also keep Dextrose ≤10% and the final solution near neutral pH • Rotate the site every 48–72 h and check the site every shift
Q11. How will you change the macronutrients to lower osmolarity?	• Approximate contribution: Dextrose 1% ≈ 50 mOsm/L; amino acids 1% ≈ 100 mOsm/L; electrolytes ≈ 2 mOsm/mEq; lipid emulsion adds little • Lower Dextrose concentration (give more of the calories as lipid) • Use a 3-in-1 (total nutrient admixture) or separate lipid piggyback: lipid emulsion is near isotonic • Reduce amino acid concentration or electrolytes where safe; increase the final volume if fluid status allows • Keep lipid ≤1 g/kg/day; monitor TG weekly
Also be ready for (exam add-ons)	• **Goals:** stop the bleeding, restore perfusion (MAP ≥65 mm Hg, lactate <2 mmol/L), correct acidosis, hypothermia and coagulopathy, prevent organ failure; nutrition without refeeding or overfeeding • **Non-pharm:** damage-control surgery, massive transfusion protocol, active warming, early enteral feeding when stable, central-line care bundle • **First-line:** blood products 1 : 1 : 1; Tranexamic acid 1 g IV then 1 g over 8 h; Norepinephrine 0.05–0.5 mcg/kg/min only as a bridge • **Pregnancy:** left uterine displacement (>20 weeks), fetal monitoring, Rh-negative → anti-D immunoglobulin; Norepinephrine and Tranexamic acid are acceptable; early obstetric input • **Renal/dialysis:** AKI from shock; avoid nephrotoxins; reduce Tranexamic acid in renal impairment; PN: restrict K, Phos, Mg and fluids • **Liver:** PN-associated liver disease: cycle PN, fish-oil lipid, lipid ≤1 g/kg/day, avoid overfeeding • **HF:** avoid volume overload and TACO with massive transfusion; concentrate PN • **GI (PUD/GERD):** stress-ulcer prophylaxis (Pantoprazole 40 mg IV daily) if ventilated >48 h or coagulopathy • **Alert:** Thiamine before Dextrose; peripheral PN ≤900 mOsm/L; Calcium replacement during massive transfusion

Case 2: TPN, Short Bowel Syndrome and CLABSI

Case details: CKD, diabetes, ulcerative colitis and short bowel syndrome. TPN to be started. Weight not given: **assume 70 kg** and state this to the examiner.

Question	Answer
Q1. What would you do before starting TPN?	• **Correct electrolytes first:** K, Mg and Phos (also Ca and Na) • Give Thiamine 100 mg IV before Dextrose • Baseline: BMP, Mg, Phos, glucose, triglycerides, LFTs, CBC, prealbumin/CRP • Start about 50% of goal calories and advance over 2–3 days if high refeeding risk • Monitor: K, Mg and Phos every 12 h for the first 3 days of feeding; glucose every 4–6 h; weight and fluid balance daily
Q2. Why is that?	• To prevent **refeeding syndrome**: insulin surge drives K, Mg and Phos into cells • Complications: hypophosphatemia, hypokalemia, arrhythmia, respiratory failure, fluid overload, thiamine-deficiency encephalopathy
Q3. Any considerations regarding his past history and TPN?	• CKD: restrict K, Phos, Mg and fluid; match protein to dialysis status; use acetate instead of chloride if acidotic • Diabetes: keep Dextrose GIR ≤4–5 mg/kg/min; add regular Insulin to the bag (about 0.05–0.1 unit per g Dextrose); glucose target 140–180 mg/dL (ICU) or 100–180 • Short bowel/UC: replace GI losses mL for mL; monitor Na, K, Mg, Zn, bicarbonate; add trace elements and vitamins • Long-term PN: use a fish-oil-containing lipid (SMOF, Omegaven) and cycle PN to protect the liver • Reassess for enteral/oral feeding (and GLP-2 analog such as Teduglutide in selected SBS)
Q4. Start 30 kcal/kg/day: how many calories?	• 70 kg × 30 = **2100 kcal/day** • Begin lower (about half) and advance if refeeding risk
Question	**Answer**
Q5. How many grams of protein and lipid, and why?	• Protein: CKD not on dialysis 0.6–0.8 g/kg = **42–56 g/day** (restricted) • If on dialysis or critically ill: 1.2–1.5 g/kg (84–105 g) to avoid negative nitrogen balance

Question	Answer
	• Lipid: maximum 1 g/kg/day = **70 g/day** (about 30% of calories); usual 0.7–1 g/kg • Example (56 g protein, 50 g lipid): protein 224 kcal + lipid 500 kcal + Dextrose 1376 kcal (405 g) = 2100 kcal • GIR = 405,000 ÷ 70 ÷ 1440 = 4.0 mg/kg/min: at the upper limit for a diabetic, so lower total calories or advance slowly
Q6. Fluids at 30 mL/kg/day: per day and per hour?	• 70 × 30 = **2100 mL/day** • 2100 ÷ 24 = **87.5 mL/h** • Add GI/ostomy losses; reduce if oliguric or overloaded (CKD)
Q7. While giving the fluids, what will you monitor?	• Daily weight, intake/output, vitals, edema and lung exam • BMP, Mg, Phos daily until stable, then 2–3 times weekly • Glucose every 4–6 h; triglycerides, LFTs weekly; CBC • Catheter site, ostomy/stool output, refeeding signs
Q8. After PN: high TG, cholesterol, LDL and ALT. Complication and management?	• Hypertriglyceridemia (lipid overload) with PN-associated liver disease (intestinal failure–associated liver disease) • Hold or cut the lipid if TG >400 mg/dL (4.5 mmol/L); keep ≤1 g/kg/day • Switch to a fish-oil-containing emulsion; avoid soybean-only lipid • Avoid overfeeding Dextrose and total calories; cycle PN over 12–16 h when stable • Add enteral/trophic feeding; treat sepsis and bacterial overgrowth; consider Ursodeoxycholic acid • Monitor: TG, LFTs, bilirubin weekly; stop hepatotoxic drugs
Q9. High fever and CLABSI. What will you do (steps)?	• Cultures: 2 sets (one peripheral, one from the catheter); differential time to positivity >2 h supports catheter source • Empiric: Vancomycin + antipseudomonal β-lactam (Cefepime 2 g IV q8h) • Add an echinocandin (Micafungin 100 mg IV daily) for catheter-related bloodstream infection with TPN and short bowel (Candida risk factors; this is a high-risk line infection, not routine sepsis) • **Remove the catheter** (mandatory for S. aureus, Pseudomonas, Candida, septic shock, persistent bacteremia) • Stop PN through the infected line; run Dextrose 10% to avoid hypoglycemia; place a new line at a new site once cultures are negative • De-escalate by culture; duration: CoNS 5–7 days, S. aureus ≥14 days (echo), gram-negatives 7–14 days, Candida 14 days after first negative culture • Monitor: temperature, WBC, repeat cultures, SCr (Vancomycin), glucose
Also be ready for (exam add-ons)	• **Goals:** meet energy and protein needs, correct fluid and electrolyte losses, avoid refeeding, liver injury and line infection • **Non-pharm:** enteral/oral intake as tolerated, oral rehydration solution, small frequent meals, CLABSI prevention bundle (hand hygiene, chlorhexidine, dedicated lumen), PN cycling • **First-line:** PN 25–30 kcal/kg; protein 0.6–0.8 g/kg (CKD, no dialysis) or 1.2–1.5 g/kg (dialysis or critical illness); lipid ≤1 g/kg; CLABSI: Vancomycin + Cefepime 2 g q8h ± Micafungin 100 mg daily • **Pregnancy:** PN needs rise (about +300 kcal/day in the 2nd–3rd trimester); folate and vitamin supplementation; β-lactams and Vancomycin are safe; avoid aminoglycosides, tetracyclines and high-dose Fluconazole; use liposomal Amphotericin B if an antifungal is essential • **Renal/dialysis:** restrict K, Phos, Mg and fluid; renal formulation; Vancomycin by AUC; Cefepime reduce (neurotoxicity); protein 1.2–1.5 g/kg once on dialysis • **Liver:** IFALD: fish-oil lipid, cycle PN, treat bacterial overgrowth, Ursodeoxycholic acid • **HF:** concentrate the PN; limit fluid; daily weights • **Counseling:** home PN: aseptic technique, daily line inspection, never use the line for other drugs or blood draws, fever or chills means call at once, glucose checks, weigh weekly

2022–2023

Case 1: Septic Shock with SBP (Vancomycin TDM, Hemodialysis, Anemia of CKD) ★ Repeated case, also in 2020, 2018–2019

Case details: SBP and encephalopathy, epilepsy, HTN, HF and DM. Admitted with SBP, then deteriorated: SBP <90 mm Hg, fever, leukocytosis. Home: Amlodipine, Furosemide, Lisinopril, Metformin, Levetiracetam, Ceftriaxone. Weight not given: assume 70 kg.

Question	Answer
Q1. Initial management, and within how long should you start it?	• **Start immediately** (treat as an emergency): resuscitation starts immediately and antimicrobials go in within 1 hour (SSC 2026) • Hold Amlodipine, Furosemide, Lisinopril and Metformin (hypotension, AKI, lactic acidosis) • Monitored bed/ICU, two large-bore IVs, lactate, cultures, source assessment (diagnostic paracentesis) • Non-pharm: monitored/ICU bed, source assessment (diagnostic paracentesis, cultures, imaging), airway and oxygen as needed • Monitor: MAP, HR, RR, SpO2, temperature, lactate (repeat in 2–4 h if >2 mmol/L), urine output, mental status
Q2. What will you do and give in the first hour (hour-1 bundle)?	• Measure lactate; remeasure if >2 mmol/L • Blood cultures before antibiotics (do not delay antibiotics >45 min) • Broad-spectrum IV antibiotics • Crystalloid 30 mL/kg (balanced crystalloid, eg Lactated Ringer, preferred over 0.9% NaCl: SSC 2026, moderate certainty) for hypotension or lactate ≥4 mmol/L • Vasopressors if hypotensive during or after fluids, to keep MAP ≥65 mm Hg • Monitor: lactate clearance, capillary refill, MAP ≥65, urine output ≥0.5 mL/kg/h; reassess volume status after each bolus (passive leg raise, pulse pressure variation)
Q3. Still hypotensive: what do you recommend?	• Norepinephrine 0.05 mcg/kg/min, titrate every 5–10 min to MAP ≥65 (usual 0.05–0.5 mcg/kg/min) • Add Vasopressin 0.03 units/min if Norepinephrine ≥0.25 mcg/kg/min • Add Hydrocortisone 50 mg IV q6h if vasopressor need persists ≥4 h at ≥0.25 mcg/kg/min • Non-pharm: central line and arterial line, continue guided fluids, source control • Monitor: MAP every 5–10 min while titrating, lactate, urine output, ECG (arrhythmia), limb/digital perfusion, glucose on Hydrocortisone
Q4. What do you think about the current medications?	• Stop all antihypertensives (Amlodipine, Lisinopril, Furosemide) and Metformin • Upgrade the antibiotic (failed Ceftriaxone, now hospital-acquired septic shock): Piperacillin-tazobactam 4.5 g IV q6h + Vancomycin • If ESBL risk or shock worsens: Meropenem 1 g IV q8h instead of Piperacillin-tazobactam • Albumin 1.5 g/kg on day 1 and 1 g/kg on day 3 for SBP with AKI (AASLD) • Continue Levetiracetam with renal adjustment • Monitor: SCr and urine output, Vancomycin AUC, cultures at 48–72 h (de-escalate), K, glucose, BP off antihypertensives; restart them stepwise after recovery
Q5. Vancomycin target?	• AUC24/MIC 400–600 mg·h/L (2020 ASHP/IDSA consensus), by Bayesian software or two levels • Older trough target 15–20 mg/L (more nephrotoxic) is no longer preferred • Loading dose 20–35 mg/kg actual weight (max 3 g)
Q6. Started hemodialysis. Instruction regarding Levetiracetam 500 mg BID?	• About 50% is removed in a 4-hour session • Give the dose after dialysis on HD days, plus a **supplemental 250–500 mg after the session** • Usual ESRD dosing: 500–1000 mg q24h
Q7. Vancomycin level now 25 (post-hemodialysis). What do you recommend?	• Hold the next dose • A level drawn right after HD is distorted by rebound; repeat 4–6 h later or pre-HD • Re-dose when the pre-HD level is <15–20 mg/L or AUC-guided (give 5–10 mg/kg after HD)
Q8. CKD with anemia of chronic disease; physician wants Erythropoietin. What do you check first?	• Iron studies: ferritin, TSAT, iron, TIBC (replace iron first if TSAT <20% or ferritin <100–200 ng/mL) • Hgb/Hct, reticulocytes, CBC; B12 and folate • Stool occult blood; infection/inflammation (CRP) • Blood pressure controlled; start only if Hgb <10 g/dL
Q9. Concerns before starting Erythropoietin?	• Causes of hyporesponse: iron/B12/folate deficiency, active infection or inflammation (this patient is septic), hyperparathyroidism, malignancy, hemolysis, malnutrition, aluminum • Safety: hypertension, stroke, MI, VTE and access thrombosis, seizures • Pure red cell aplasia (anti-erythropoietin antibodies)
Q10. Monitoring parameters?	• Hgb every 2–4 weeks at start, then monthly (dialysis) or every 3 months (non-dialysis) • Target Hgb 10–11.5 g/dL; reduce or hold if rise >1 g/dL in 2 weeks or Hgb >11.5 • Blood pressure, iron indices every 3 months, K and chemistry • Thrombotic events, seizure, response failure
Q11. Weight-based or fixed dose?	• Weight-based: Epoetin alfa 50–100 units/kg IV/SC 3 times weekly • Darbepoetin 0.45 mcg/kg weekly
Q12. On which weight?	• Actual (dry) body weight

Question	Answer
Also be ready for (exam add-ons)	• **Goals:** MAP ≥65, lactate clearance, control the source, prevent organ failure, avoid drug toxicity • **Non-pharm:** source control (paracentesis, drainage, remove devices), early mobilization, head of bed 30°, VTE prophylaxis (Enoxaparin or devices), glucose 140–180 mg/dL • **First-line:** Norepinephrine 0.05 mcg/kg/min, balanced crystalloid 30 mL/kg, Piperacillin-tazobactam 4.5 g q6h (or Meropenem 1 g q8h) + Vancomycin; Albumin for SBP • **Pregnancy:** same hour-1 bundle; left lateral position; Norepinephrine is acceptable; β-lactams and Vancomycin are safe; avoid aminoglycosides, tetracyclines, fluoroquinolones • **Renal/dialysis:** Piperacillin-tazobactam in HD: 2.25 g q12h (extra 0.75 g after HD per label); Meropenem CrCl 10–25: 500 mg q12h; Levetiracetam supplement after HD; Vancomycin AUC-based, redose by pre-HD level • **Liver:** SBP with cirrhosis: Albumin 1.5 g/kg day 1 and 1 g/kg day 3; avoid NSAIDs and aminoglycosides; Hydrocortisone no adjustment • **HF:** reassess volume after each bolus; Dobutamine only if cardiac dysfunction with hypoperfusion; hold ACE inhibitor and β-blocker during shock • **GI (PUD/GERD):** stress-ulcer prophylaxis (PPI) in shock, coagulopathy or ventilation • **Counseling:** before discharge review all antihypertensives, Metformin and diuretics (restart stepwise); vaccines; EPO: BP and thrombosis

2021

Case 1: TDM of Vancomycin and Gentamicin (Septic Shock, Febrile Neutropenia) ★ Repeated case, also in 2020, 2018–2019

Case details: Septic shock (or febrile neutropenia). Weight, height and SCr are missing: ask the nurse. SCr is reported in µmol/L, so convert to mg/dL before calculating CrCl.

Question	Answer
Q1. How do you calculate CrCl?	• Convert: SCr mg/dL = µmol/L ÷ 88.4 (106 ÷ 88.4 = **1.2 mg/dL**) • Cockcroft-Gault: CrCl = (140 − age) × weight ÷ (72 × SCr), × 0.85 in women • Weight: IBW if normal weight; adjusted weight (IBW + 0.4 × [actual − IBW]) if >120–130% of IBW; actual if underweight • Example: (140 − 60) × 80 ÷ (72 × 1.2) = **74 mL/min**
Q2. Vancomycin: loading dose, maintenance dose, target and level timing?	• Loading: 25 mg/kg actual weight (range 20–35, max 3 g) = 2000 mg (example) • Maintenance: 15 mg/kg = 1200 mg, round to 1250 mg q12h (CrCl 50–90: q12h; >90: q8h; 15–50: q24h) • Target: AUC24/MIC 400–600 mg·h/L (two levels or Bayesian); trough 15–20 is the older surrogate • Level timing: trough before the 4th dose at steady state; for AUC take a peak 1–2 h after the end of infusion and a trough • Monitor: SCr daily, urine output, avoid other nephrotoxins
Q3. Gentamicin dose (by IBW), target and timing?	• Dosing weight: IBW (70.5 kg); use adjusted weight if actual >120% of IBW • Extended interval: 5–7 mg/kg q24h = 7 × 70.5 = about 490 mg q24h (CrCl >60); q36h if CrCl 40–59, q48h if 20–39 (Hartford) • Conventional alternative: 1.5–2 mg/kg load, then 1–1.7 mg/kg q8h • Level: 6–14 h after the first dose on the Hartford nomogram; or peak 30 min after the end of a 30-min infusion • Targets: extended-interval trough <1 mg/L (undetectable); conventional peak 5–10, trough <2 mg/L • Monitor: SCr, urine output, hearing/balance, K and Mg
Q4. Febrile neutropenia: empiric antibiotics (2018–2019)?	• Cefepime 2 g IV q8h (or Piperacillin-tazobactam 4.5 g q6h or Meropenem 1 g q8h) within 1 hour • Add Vancomycin only for hemodynamic instability, catheter infection, skin/soft tissue, pneumonia or MRSA colonization • Add an aminoglycoside (Gentamicin or Amikacin) if shock or resistant gram-negative risk • Persistent fever 4–7 days: empiric antifungal (Micafungin or Caspofungin) is accepted in neutropenia (IDSA/ASCO), unlike routine sepsis; reassess CT chest • Stop after ANC >500 and afebrile 48 h • Non-pharm: neutropenic precautions, hand hygiene, remove/evaluate central lines, daily skin and mouth check • Monitor: temperature every 4 h, ANC and CBC daily, cultures, SCr (aminoglycoside/Vancomycin), CRP, chest imaging if persistent fever
Question	**Answer**
Also be ready for (exam add-ons)	• **Goals:** reach the target exposure early (Vancomycin AUC 400–600; Gentamicin peak/trough by regimen) with minimal nephro- and ototoxicity • **Non-pharm:** hydration, avoid other nephrotoxins (NSAIDs, contrast, Piperacillin-tazobactam with Vancomycin when possible), daily SCr and urine output

Question	Answer
	• **First-line:** Vancomycin load 25 mg/kg, maintenance 15 mg/kg q8–24h by CrCl; Gentamicin 5–7 mg/kg q24h (IBW or adjusted weight); febrile neutropenia: Cefepime 2 g q8h • **Pregnancy:** Vancomycin clearance rises (higher or more frequent doses, monitor levels); Gentamicin only for life-threatening infection (fetal ototoxicity); β-lactams preferred • **Renal/dialysis:** HD: Vancomycin load 25 mg/kg, redose 500–1000 mg after HD when the pre-HD level is below 15–20; Gentamicin 1.5–2 mg/kg after HD with level; CRRT: higher doses • **Liver:** no dose change; cirrhosis with ascites raises Vd (use actual weight for the load) • **HF/obesity:** larger Vd (load by actual weight); Gentamicin by adjusted weight if >120% of IBW • **Counseling:** infuse Vancomycin ≤1 g/h (infusion reaction/red man syndrome); report hearing loss, tinnitus, dizziness, low urine output • **Alert:** Vancomycin + Piperacillin-tazobactam raises the AKI rate

Note: Worked example (assumed values, state them aloud): 60-year-old man, 80 kg, height 175 cm (IBW 70.5 kg), SCr 106 µmol/L.

2020

Case 1: Sepsis with Intra-abdominal Infection (qSOFA, Vasopressors, MDR Klebsiella, Hydrocortisone) ★ Repeated case, also in 2018–2019

Case details: Intra-abdominal surgery, then sepsis. Started on Meropenem + Vancomycin. Hemodynamically unstable despite fluids and Norepinephrine. Culture: Klebsiella pneumoniae resistant to all cephalosporins and Ampicillin-sulbactam.

Question	Answer
Q1. What is the qSOFA and why do we use it?	• Quick bedside score, 1 point each: respiratory rate **≥22**, systolic BP **≤100 mm Hg**, altered mentation (GCS <15) • Score ≥2 = suspected infection with higher risk of death or prolonged ICU stay (Sepsis-3) • Use: screen outside the ICU; it is a prompt, not a diagnosis; SSC 2026 (strong recommendation) says to screen with NEWS/NEWS2, MEWS or SIRS rather than qSOFA alone • Sepsis = infection + organ dysfunction (SOFA rise ≥2); septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L despite fluids
Q2. The team wants Albumin as resuscitation fluid. Your opinion?	• First-line: crystalloid, preferably balanced (Lactated Ringer or Plasma-Lyte) over 0.9% NaCl • Initial volume: at least 30 mL/kg within the first 3 hours (SSC 2026, low certainty; use ideal or adjusted weight if BMI >30); 15 mL/kg is too little • Albumin: SSC 2026 suggests crystalloids alone over crystalloids + albumin (moderate certainty); consider albumin only after very large crystalloid volumes or in cirrhosis, and avoid it in traumatic brain injury • Dose reference (ALBIOS): 20% Albumin 20–40 g/day to keep serum albumin ≥3 g/dL • Avoid starch and gelatin
Q3. Patient got 2400 mL and Norepinephrine; BP 80/40. What now?	• Norepinephrine: titrate (typical 0.05–0.5 mcg/kg/min; 80 kg × 0.1 = 8 mcg/min) • Add **Vasopressin 0.03 units/min** (fixed dose) when Norepinephrine is about 0.25–0.5 mcg/kg/min • Third agent: Epinephrine 0.05–0.5 mcg/kg/min • Dobutamine 2.5–20 mcg/kg/min only if persistent hypoperfusion with cardiac dysfunction despite MAP and volume • Continue fluids guided by dynamic measures (passive leg raise, pulse pressure variation) and capillary refill • Monitor: MAP every 5 min, lactate every 2–4 h, urine output, ScvO2/capillary refill, ECG, limb perfusion
Q4. When to give Hydrocortisone, and when to stop it?	• SSC 2026 (low certainty) suggests IV corticosteroids in septic shock with an ongoing vasopressor requirement: Hydrocortisone 200 mg/day (50 mg q6h or infusion); the 2021 trigger was Norepinephrine ≥0.25 mcg/kg/min for ≥4 h (confirm the exact 2026 wording in the full text) • Do not increase the dose if still unstable (no extra benefit above 200–400 mg/day) • Stop/taper once vasopressors are off; the usual course is about 3–7 days. Continuing for 2 weeks after stabilization is not supported • Not for sepsis without shock • Monitor: glucose (144–180 mg/dL), sodium, myopathy, delirium, GI bleeding • Note: SSC 2016 advised against steroids if fluids and pressors restore stability; SSC 2021 and 2026 are more permissive for ongoing vasopressor need
Question	**Answer**
Q5. Culture: Klebsiella resistant to all cephalosporins, still unstable. Your action?	• Stop Vancomycin: no gram-positive pathogen, nephrotoxic • Confirm carbapenem susceptibility (MIC): if ESBL only, Meropenem 1 g IV q8h (extended infusion) is the drug of choice • If carbapenem-resistant (CRE) with failure on Meropenem (exam answer): add **Colistin** 9 million IU load, then 4.5 million IU q12h (renal adjustment), plus Meropenem 2 g q8h extended infusion

Question	Answer
	• Guideline-preferred if available (IDSA 2024): Ceftazidime-avibactam 2.5 g IV q8h (KPC), Meropenem-vaborbactam 4 g q8h or Imipenem-relebactam; for metallo-β-lactamase use Ceftazidime-avibactam + Aztreonam or Cefiderocol • Source control; duration 4–7 days after adequate source control • Monitor: SCr (Colistin nephrotoxicity), neurotoxicity, cultures
Q6. Patient stabilized. Should he continue the Probiotic for C. difficile?	• No routine Probiotic: insufficient evidence for treatment or prevention (IDSA/SHEA 2021); fungemia/bacteremia risk in critically ill with central lines • Prevent: stop unnecessary antibiotics/PPIs, hand hygiene with soap and water • If CDI occurs: Fidaxomicin 200 mg PO BID × 10 days (preferred) or Vancomycin 125 mg PO q6h × 10 days
Q7. When to start an antifungal?	• Not routine in sepsis • **SSC 2026 suggests against empiric antifungal therapy** (low certainty); reserve an echinocandin (Micafungin 100 mg IV daily) for selected very-high-risk patients per the IDSA candidiasis guideline, with risk factors: neutropenia, TPN, anastomotic leak/recurrent perforation, prolonged broad-spectrum antibiotics, central line, Candida colonization at multiple sites, immunosuppression
Q8. For how long should antibiotics continue?	• 7–10 days for most serious infections; 4 days after adequate source control in intra-abdominal infection (STOP-IT) • Longer: slow response, undrained focus, S. aureus bacteremia, some fungal/viral, neutropenia/immunodeficiency • Shorter: rapid resolution after source control, uncomplicated pyelonephritis • Daily de-escalation; Procalcitonin may support stopping; Vancomycin stops when MRSA is excluded (cultures negative at 48–72 h)
Also be ready for (exam add-ons)	• **Goals:** hemodynamic stability (MAP ≥65, lactate normal), source control, targeted therapy, de-escalation, avoid steroid and drug toxicity • **Non-pharm:** source control (drainage, surgery), balanced crystalloid, early mobilization, VTE prophylaxis, glucose control, hand hygiene to prevent CDI • **First-line:** Norepinephrine; add Vasopressin 0.03 units/min; Hydrocortisone 200 mg/day; empiric Piperacillin-tazobactam or Meropenem; CRE: Ceftazidime-avibactam 2.5 g q8h • **Pregnancy:** avoid Colistin unless no alternative; avoid Tigecycline and aminoglycosides; Meropenem and Piperacillin-tazobactam acceptable; left lateral positioning • **Renal/dialysis:** Colistin is nephrotoxic (load regardless of renal function, then adjust); Ceftazidime-avibactam adjust to CrCl; Meropenem adjust • **Liver:** Tigecycline reduce dose in Child-Pugh C; hepatotoxic drugs avoided; steroids no adjustment • **HF:** Dobutamine for low cardiac output; avoid over-resuscitation • **GI (PUD/GERD):** PPI stress-ulcer prophylaxis; C. difficile: Fidaxomicin 200 mg BID × 10 days or Vancomycin 125 mg q6h • **Counseling:** steroid effects (hyperglycemia, myopathy, delirium); report diarrhea after antibiotics

2015

Case 1: Critical Care Long Case (Sepsis, Acid-Base Disorder, AKI)

Case details: Long case combining sepsis, an acid-base disorder and AKI. Questions typically ask for vasopressors, inotropes and antibiotics (differences, advantages/disadvantages, alternatives) with justification.

Question	Answer
Q1. Compare the vasopressors and inotropes.	• **Norepinephrine:** first line; α1 > β1; raises MAP with little tachycardia; risk of digital/gut ischemia • Vasopressin: V1 vasoconstriction; spares catecholamine; fixed 0.03 units/min; risk of digital/mesenteric ischemia; do not start as first agent • Epinephrine: α + β1/β2; second/third line; raises lactate, tachyarrhythmia, hyperglycemia • Phenylephrine: pure α1; use for tachyarrhythmia; reflex bradycardia, lowers cardiac output • Dopamine: avoid (arrhythmia); only in selected bradycardia • Dobutamine: β1 inotrope 2.5–20 mcg/kg/min for cardiac dysfunction with hypoperfusion; hypotension, arrhythmia • Alternative: Angiotensin II 20 ng/kg/min for refractory shock; Methylene blue in selected vasoplegia • Monitor: MAP every 5 min, arterial line, ECG and HR, lactate, urine output, cardiac output/ScvO2 when an inotrope is used, extremity perfusion
Question	**Answer**
Q2. Sepsis antibiotics: how will you choose and justify?	• Within 1 hour in shock; cover likely source and local resistance; two gram-negative agents for high-risk MDR, MRSA coverage if risk • Obtain cultures first; de-escalate at 48–72 h; renal dosing; extended-infusion β-lactams • Source control within 6–12 h • Non-pharm: source control (drainage, remove infected devices) within 6–12 h

Question	Answer
	• Monitor: temperature, WBC, procalcitonin, cultures at 48–72 h, SCr and drug levels (Vancomycin AUC, aminoglycoside), clinical stability for de-escalation
Q3. How do you approach the acid-base disorder?	• Step 1: pH (acidemia/alkalemia); Step 2: PaCO2 (respiratory) and HCO3 (metabolic) • Step 3: anion gap = Na − (Cl + HCO3); correct for albumin (+2.5 per 1 g/dL below 4) • Step 4: compensation (Winter formula); delta-delta for mixed disorders • Common: lactic acidosis in sepsis (treat perfusion), hyperchloremic acidosis (stop saline), AKI acidosis • Bicarbonate: only pH ≤7.2 with AKI stage 2–3 (BICAR-ICU); avoid in lactic acidosis with higher pH
Q4. How do you classify and manage the AKI?	• KDIGO stage 1: SCr ≥0.3 mg/dL rise or 1.5–1.9× baseline; stage 2: 2.0–2.9×; stage 3: ≥3×, SCr ≥4.0, or RRT • Urine output: <0.5 mL/kg/h × 6–12 h (1), ≥12 h (2), <0.3 × 24 h or anuria 12 h (3) • Prerenal: FeNa <1%; ATN: >2% (FeUrea <35% prerenal) • Manage: restore MAP ≥65 and volume (balanced crystalloid), stop nephrotoxins (NSAIDs, ACE inhibitors, contrast, Vancomycin where possible), renally adjust drugs • RRT indications: refractory hyperkalemia, acidosis (pH <7.1), fluid overload, uremic complications, dialyzable toxin • Monitor: SCr, urine output, K, HCO3, drug levels
Also be ready for (exam add-ons)	• **Goals:** restore perfusion and oxygen delivery, correct acid-base status, protect the kidney, treat the infection • **Non-pharm:** source control, balanced fluids, avoid nephrotoxins, early mobilization, nutrition • **First-line:** Norepinephrine, then Vasopressin; empiric broad-spectrum antibiotics within 1 h; Bicarbonate only if pH ≤7.2 with AKI • **Pregnancy:** left lateral positioning; Norepinephrine acceptable; avoid aminoglycosides and fluoroquinolones • **Renal/dialysis:** RRT for refractory hyperkalemia, acidosis, fluid overload, uremia; renally adjust all drugs • **Liver:** lactate clearance is reduced in liver failure; avoid hepatotoxic drugs • **HF:** inotrope (Dobutamine) for cardiogenic component; careful fluid challenge

SOE Bank (year not stated)

Case 1: Septic Shock with Pneumonia (Mechanically Ventilated)

Case details: 68-year-old woman with pneumonia, BP 60/40 (MAP 46), admitted to ICU and mechanically ventilated. Weight 70 kg, SCr 1.2 mg/dL, glucose 12 mmol/L. Height is not given: **assume 160 cm** (IBW 52 kg) and say so.

Question	Answer
Q1. What type of fluid will you choose, and at what rate?	• **Balanced crystalloid** (Lactated Ringer or Plasma-Lyte) over 0.9% NaCl (SSC 2026, moderate certainty); use 0.9% NaCl only if hypochloremic or hyponatremic (the sodium hint) • Volume: **at least 30 mL/kg within the first 3 hours** (SSC 2026, low certainty; ideal or adjusted weight if BMI >30) = about 2100 mL here • Give as 500 mL boluses over 15–30 min and reassess after each (dynamic measures: passive leg raise, pulse-pressure or stroke-volume variation) • Albumin: SSC 2026 suggests crystalloids alone over adding albumin (moderate certainty); consider only after very large crystalloid volumes or in cirrhosis • MAP 46: start Norepinephrine early through a peripheral line; do not wait for the full fluid volume
Q2. Hemodynamic goals (at least 3)?	• **MAP ≥65 mm Hg** (SSC 2026: initial target; 60–65 is acceptable in patients ≥65 years, as here) • Lactate falling toward normal (repeat every 2–4 h while >2 mmol/L) • Capillary refill time ≤3 s, mental status, skin mottling, urine output ≥0.5 mL/kg/h • Dynamic parameters (passive leg raise, SV/SVV, PPV, echocardiography) rather than static CVP • CVP 8–12 mm Hg and ScvO2 >70% are legacy early-goal-directed targets: not required in current guidelines
Q3. Which antibiotics will you choose, and why? (CrCl, doses)	• Within **1 hour** of recognition (SSC 2026, strong); prolonged (extended) β-lactam infusion after the first dose (strong) • Ventilated pneumonia in shock: cover Streptococcus pneumoniae, atypicals and gram-negatives including Pseudomonas: Piperacillin-tazobactam 4.5 g IV q6h (no adjustment if CrCl >40) • Add MRSA coverage if risk factors: Vancomycin 25 mg/kg load = 1750 mg, then about 1250 mg q24h at CrCl ≈46–50 mL/min, by AUC 400–600 • Atypical cover: Azithromycin 500 mg IV daily (check QTc) • CrCl (Cockcroft-Gault, female) = (140 − 68) × 70 ÷ (72 × 1.2) × 0.85 = **about 50 mL/min** (the answer key says 46 using another weight) • SSC 2026: no routine anaerobic coverage unless a risk factor exists (Piperacillin-tazobactam is acceptable anyway)

Question	Answer
Q4. Still hypotensive despite fluids and antibiotics: what do you add? Safety and efficacy?	• **Norepinephrine first line:** start 0.05–0.1 mcg/kg/min, titrate every 5–10 min to MAP goal (usual range up to 0.5; the answer key's start of 0.25–0.5 is a high-dose range, not a starting dose) • Add **Vasopressin 0.03 units/min** at about 0.25–0.5 mcg/kg/min of Norepinephrine to raise MAP or spare Norepinephrine • Add Epinephrine if MAP is still inadequate on both (SSC 2026, very low certainty); Dobutamine only for persistent shock with cardiac dysfunction • Efficacy: MAP, lactate, urine output, capillary refill. Safety: tachyarrhythmia (ECG), limb/digital and gut ischemia, extravasation (central line), arterial line for intermediate-to-high doses
Q5. Is she at risk of MRSA? What agent? Duration?	• Risk: prior MRSA isolation or colonization, IV antibiotics or hospitalization in the last 90 days, ventilation, dialysis, skin infection • Send a **nasal MRSA PCR**: a negative result lets you stop Vancomycin (high negative predictive value) • Vancomycin AUC 400–600 mg·h/L (trough 15–20 is the older surrogate); first level before the 3rd–4th dose • Duration: 7 days for good responders (HAP/VAP); stop earlier if cultures and PCR are negative
Q6. Indications for corticosteroids? Agent and dose?	• SSC 2026 (low certainty): suggests IV corticosteroids in septic shock with an ongoing vasopressor requirement • Agent: **Hydrocortisone 200 mg/day** (50 mg q6h or continuous infusion); 2021 trigger was Norepinephrine ≥0.25 mcg/kg/min for ≥4 h (confirm the exact 2026 wording) • Stop (taper) once vasopressors are off; monitor glucose and sodium
Q7. Which stress-ulcer prophylaxis, and why?	• High risk: mechanical ventilation >48 h, shock and sepsis (and coagulopathy) • Pantoprazole 40 mg IV daily (or Omeprazole 40 mg); stop when the risk factors resolve
Q8. Which DVT prophylaxis, and why?	• Enoxaparin 40 mg SC once daily (CrCl ≥30); LMWH is suggested over UFH (lower DVT rate) • CrCl <30: UFH 5000 units SC q8–12h; contraindication to anticoagulants: intermittent pneumatic compression
Q9. Hyperglycemia: agent, monitoring, target?	• Glucose 12 mmol/L (216 mg/dL) is above the 10 mmol/L (180 mg/dL) trigger: start **IV regular Insulin** (about 1–4 units/h by protocol) • Target 8–10 mmol/L (144–180 mg/dL); avoid <6 mmol/L (hypoglycemia) • Monitor glucose hourly until stable, then every 2–4 h; K with Insulin
Q10. Which home drugs will you stop, and why?	• All antihypertensives and diuretics (shock); Metformin (lactic acidosis, AKI); NSAIDs and ACE inhibitors/ARBs (AKI); SGLT2 inhibitors (euglycemic DKA, volume depletion) • Restart stepwise as shock and kidney function recover
Q11. Calculate CrCl and the causes of AKI.	• CrCl about 50 mL/min (see Q3); AKI causes: septic hypoperfusion, nephrotoxins (NSAIDs, ACE inhibitors/ARBs, Vancomycin + Piperacillin-tazobactam together), contrast, diuretics
Q12. Acid-base 1: pH 7.48, PCO2 50, HCO3 28. Type and treatment?	• High pH and high HCO3 = **metabolic alkalosis** • Expected PCO2 = 40 + 0.7 × (28 − 24) ≈ 43; measured 50 is higher: added respiratory acidosis or hypoventilation (check ventilator settings) • Urine chloride <25 mEq/L (saline-responsive: volume contraction, diuretics, NG suction): 0.9% NaCl and K repletion • Urine chloride >25 mEq/L (saline-resistant: hyperaldosteronism, severe hypokalemia): correct K, Spironolactone 12.5–25 mg; Acetazolamide 250–500 mg IV if volume overloaded
Q13. Acid-base 2: Na 140, Cl 100, pH 7.30, PCO2 30, HCO3 20.	• Anion gap = 140 − (100 + 20) = **20** (high): high anion-gap metabolic acidosis (lactate, ketones, uremia, toxins) • Winter: expected PCO2 = 1.5 × 20 + 8 = 38 ± 2; measured 30 is lower: concomitant respiratory alkalosis (sepsis tachypnea) • Delta ratio = (20 − 12) ÷ (24 − 20) = 2: pure or borderline mixed with metabolic alkalosis • Treat the cause (perfusion for lactate, Insulin for DKA); Bicarbonate only if pH ≤7.2 with AKI stage 2–3 (BICAR-ICU)
Question	**Answer**
Also be ready for (exam add-ons)	• **Goals:** MAP ≥65 (60–65 acceptable if ≥65 years), lactate clearance, eradicate the pathogen, avoid ventilator and drug complications, de-escalate early • **Non-pharm:** lung-protective ventilation (6 mL/kg IBW), head of bed 30–45°, daily sedation interruption and spontaneous breathing trials, oral care, early mobilization, VTE and stress-ulcer prophylaxis, glucose control • **First-line:** balanced crystalloid ≥30 mL/kg in 3 h; Norepinephrine → add Vasopressin; Piperacillin-tazobactam 4.5 g q6h (extended infusion) ± Vancomycin (nasal MRSA PCR) ± Azithromycin; Hydrocortisone 200 mg/day if vasopressors persist • **Pregnancy:** same bundle with left lateral positioning; Norepinephrine acceptable; β-lactams and Vancomycin safe; avoid fluoroquinolones, tetracyclines and aminoglycosides; pregnancy-adjusted ventilation and oxygenation targets • **Renal/dialysis:** Piperacillin-tazobactam 2.25 g q8h (CrCl <20) or q12h in HD; Vancomycin by AUC with levels after HD; Meropenem if switched: adjust to CrCl

Question	Answer
	• **Liver:** Hydrocortisone no adjustment; avoid hepatotoxic combinations; Azithromycin caution in severe disease • **HF:** avoid fluid overload after resuscitation (SSC 2026 suggests active fluid removal later); Dobutamine only for cardiac dysfunction with hypoperfusion • **GI (PUD/GERD):** PPI stress-ulcer prophylaxis (ventilation >48 h, shock, coagulopathy); avoid NSAIDs • **Counseling:** (ICU) family communication; at discharge: vaccines (pneumococcal, influenza, COVID-19, RSV), smoking cessation, follow-up chest imaging when indicated

Case 2: Septic Shock with Recurrent UTI

> **Case details:** 38-year-old woman with a UTI that recurred within 14 days of stopping antibiotics. Febrile, BP 60/40, GCS 12.

Question	Answer
Q1. Calculate the qSOFA. When can we use it?	• qSOFA = **3** if RR ≥22 (low SBP ≤100 + GCS <15 + RR); the stem gives two criteria and the answer key counts 3 • Use outside the ICU as a prompt for risk of death or prolonged ICU stay • **SSC 2026 (strong):** screen with NEWS/NEWS2, MEWS or SIRS rather than qSOFA alone
Q2. When do you start an antifungal?	• SSC 2026 suggests against routine empiric antifungals (low certainty) • Consider only with strong risk factors: recent abdominal surgery or anastomotic leak, long-term TPN, central line, immunocompromise (neutropenia, transplant, prolonged steroids), Candida colonization at multiple sites, or deterioration despite adequate antibiotics • Agent: Micafungin 100 mg IV daily (echinocandin) in critically ill patients
Q3. When to give Hydrocortisone, and when to stop it?	• Septic shock with an ongoing vasopressor requirement (see the first case): Hydrocortisone 200 mg/day (50 mg q6h) • Stop (taper) once vasopressors are no longer needed
Q4. If the patient is unstable, can we increase the Hydrocortisone dose?	• **No:** 200 mg/day is the studied dose (more does not help) • Add Vasopressin 0.03 units/min; if Norepinephrine is at its maximum add Epinephrine
Q5. When to stop Vancomycin?	• UTI source rarely needs MRSA cover: stop when cultures show no MRSA at 48–72 h • If MRSA is documented complete 7–10 days (de-escalate once the organism is known: strong recommendation in SSC 2026)
Q6. De-escalate with the culture and susceptibility results.	• Narrowest active agent: susceptible E. coli → Ceftriaxone; ESBL → Meropenem 1 g q8h; step down orally (TMP-SMX or a fluoroquinolone if susceptible) once stable
Q7. What is the duration of antibiotics?	• Same organism within 2 weeks = **relapse**: look for a source (obstruction, stone, abscess, prostatitis in men, retention) with renal ultrasound or CT and drain it first • Urosepsis: 7–14 days if source controlled and rapid response; extend to 2–6 weeks only for abscess, prostatitis or an undrained focus • Consider suppression or prophylaxis according to the cause and the patient's status
Also be ready for (exam add-ons)	• **Goals:** hemodynamic stability, source control, targeted antibiotic with de-escalation, prevent relapse • **Non-pharm:** urgent imaging and drainage of obstruction (stent or nephrostomy), remove or change catheters, fluids, hydration, early mobilization, VTE prophylaxis • **First-line:** Meropenem 1 g IV q8h (extended infusion) or Piperacillin-tazobactam 4.5 g q6h for suspected ESBL/recurrent resistant UTI; Norepinephrine for shock; Hydrocortisone 200 mg/day if ongoing vasopressor need • **Pregnancy:** pyelonephritis/urosepsis in pregnancy: Ceftriaxone or Meropenem acceptable; avoid fluoroquinolones, TMP-SMX and aminoglycosides; fetal monitoring; left lateral position • **Renal/dialysis:** Meropenem CrCl 26–50: 1 g q12h; 10–25: 500 mg q12h; Fosfomycin/Nitrofurantoin are not for urosepsis • **Liver:** no major adjustment • **Counseling:** complete the course, hydration, timed voiding, treat early symptoms, culture before antibiotics, consider prophylaxis only after source evaluation; DM: SGLT2 inhibitors raise genital-infection risk

Case 3: Septic Shock, TDM Worked Example (Vancomycin and Gentamicin)

Case details: 60-year-old man, RR 30, HR 150, T 39 °C, mental status change, BP 50/60. SCr 0.7 mg/dL, weight 60 kg, height 168 cm.

Question	Answer
Q1. Calculate CrCl (Cockcroft-Gault).	• (140 − 60) × 60 ÷ (72 × 0.7) = 4800 ÷ 50.4 = **95 mL/min** • Actual weight 60 kg is below IBW (64.2 kg), so use the actual weight
Q2. Vancomycin plan?	• Load 25 mg/kg × 60 = **1500 mg** • Maintenance 15–20 mg/kg q8–12h at CrCl 95: for example 1000–1250 mg q12h (the answer key: 1250 mg BID) • Target **AUC24 400–600 mg·h/L** (two levels or Bayesian software); trough 15–20 is the older method: level before the 4th dose
Q3. Gentamicin plan?	• IBW = 50 + 0.91 × (168 − 152.4) = 64.2 kg; **actual weight (60 kg) is lower, so dose on 60 kg** (the answer key used IBW) • Extended interval: Hartford 7 mg/kg × 60 = **420 mg q24h** (Urban-Craig 5 mg/kg = 300 mg) • Level: random level 6–14 h after the start, plotted on the nomogram; or trough before the next dose (<1 mg/L) • Traditional: load 2 mg/kg = 120 mg, then 1.7 mg/kg = 100 mg q8h; trough <1–2 mg/L, peak 5–10 mg/L • In septic shock use an aminoglycoside only for 24–72 h (empiric), then stop or de-escalate
Q4. Monitoring parameters and frequency for the aminoglycoside?	• SCr at least daily (through day 5), urine output; trough weekly if renal function is stable • Ototoxicity and vestibular toxicity: tinnitus, hearing loss, gait disturbance
Also be ready for (exam add-ons)	• **Goals:** reach the target exposure early (Vancomycin AUC 400–600; Gentamicin peak/trough by regimen) with minimal nephro- and ototoxicity • **Non-pharm:** hydration, avoid other nephrotoxins (NSAIDs, contrast, Piperacillin-tazobactam with Vancomycin when possible), daily SCr and urine output • **First-line:** Vancomycin load 25 mg/kg, maintenance 15 mg/kg q8–24h by CrCl; Gentamicin 5–7 mg/kg q24h (IBW or adjusted weight); febrile neutropenia: Cefepime 2 g q8h • **Pregnancy:** Vancomycin clearance rises (higher or more frequent doses, monitor levels); Gentamicin only for life-threatening infection (fetal ototoxicity); β-lactams preferred • **Renal/dialysis:** HD: Vancomycin load 25 mg/kg, redose 500–1000 mg after HD when the pre-HD level is below 15–20; Gentamicin 1.5–2 mg/kg after HD with level; CRRT: higher doses • **Liver:** no dose change; cirrhosis with ascites raises Vd (use actual weight for the load) • **HF/obesity:** larger Vd (load by actual weight); Gentamicin by adjusted weight if >120% of IBW • **Counseling:** infuse Vancomycin ≤1 g/h (infusion reaction/red man syndrome); report hearing loss, tinnitus, dizziness, low urine output • **Alert:** Vancomycin + Piperacillin-tazobactam raises the AKI rate